

READING PASSAGE 1 You should spend about 20 minutes on Questions 1-13, which are based on Reading Passage 1 below.

The Evolution of Smart Cities **A.** The concept of a 'smart city' has evolved significantly over the last decade. Initially, the focus was entirely on implementing new technologies such as sensor networks and automated traffic lights. City planners believed that simply installing hardware would automatically resolve urban issues like congestion and pollution. **B.** However, recent studies suggest that technology alone is insufficient. Human capital and community engagement are now recognized as equally critical. For instance, Barcelona's smart city initiative succeeded largely because it created open-source platforms where citizens could propose solutions to neighborhood problems. **C.** Despite these advancements, funding remains a significant hurdle. Upgrading legacy infrastructure to support massive data collection requires billions of dollars. Many local governments are turning to public-private partnerships to share the financial burden and technological risks.

Questions 1-4 Do the following statements agree with the information given in Reading Passage 1? Write: **TRUE** if the statement agrees with the information **FALSE** if the statement contradicts the information **NOT GIVEN** if there is no information on this

1. Early smart city projects focused mainly on human community engagement.
2. Barcelona's initiative involved citizens in solving local problems.
3. Upgrading old city infrastructure is a cheap process.
4. Public-private partnerships have successfully eliminated all technological risks in modern cities.

Questions 5-8 Complete the sentences below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

5. In the beginning, planners thought that installing _____ would solve urban problems.
6. Modern research indicates that _____ is just as important as technology.
7. Barcelona utilized _____ to allow citizens to suggest ideas.
8. To solve financial problems, governments are forming _____ with private companies.

READING PASSAGE 2 You should spend about 20 minutes on Questions 9-22, which are based on Reading Passage 2 below.

Deep Ocean Exploration **A.** The deep ocean remains one of the least explored environments on Earth. Covering more than 60% of the planet's surface, the abyssal zone is characterized by extreme pressure, total darkness, and freezing temperatures. **B.** Historically, scientists relied on sonar to map the ocean floor. While useful, sonar only provides low-resolution topographical maps. It wasn't until the development of Remotely Operated Vehicles (ROVs) in the 1980s that high-definition video and physical sampling became possible. **C.** Recent ROV expeditions have discovered unique ecosystems built around hydrothermal vents. These ecosystems do not rely on

sunlight. Instead, bacteria use a process called chemosynthesis to convert toxic minerals into usable energy, forming the base of a bizarre food web.

Questions 9-11 Reading Passage 2 has three paragraphs, A-C. Which paragraph contains the following information? Write the correct letter, A, B, or C.

9. An explanation of how energy is produced without sunlight.
10. A description of the harsh physical conditions of the deep sea.
11. The limitation of early mapping technologies.

Questions 12-13 Choose TWO letters, A-E. Which TWO of the following are mentioned as characteristics of the abyssal zone?

- A. High levels of sunlight B. Extreme pressure C. Abundant plant life D. Freezing temperatures
E. Shallow waters

Questions 14-16 Choose the correct letter, A, B, C or D.

14. What was the main advantage of ROVs developed in the 1980s? A. They were cheaper than sonar. B. They could carry human crews. C. They provided high-definition video and samples. D. They could operate without electricity.
15. Hydrothermal vent ecosystems are unique because they: A. Are found near the surface. B. Do not require sunlight to survive. C. Consist entirely of bacteria. D. Are rapidly disappearing.
16. The base of the food web near hydrothermal vents is formed by: A. Small fish B. Seaweed C. ROVs D. Bacteria

READING PASSAGE 3 You should spend about 20 minutes on Questions 17-25, which are based on Reading Passage 3 below.

The Psychology of Language Acquisition Understanding how humans acquire language has been a central debate in linguistics for decades. Behaviorists, like B.F. Skinner, argued that language is learned through imitation and reinforcement. If a child says 'milk' and receives milk, the behavior is reinforced. However, Noam Chomsky challenged this, proposing that humans are born with an innate biological capacity for language, which he called the Language Acquisition Device (LAD). Chomsky noted that children often generate sentences they have never heard before, which cannot be explained by simple imitation. Today, most cognitive scientists agree that language acquisition is a complex interaction between innate biological predispositions and rich environmental input.

Questions 17-21 Complete the summary using the list of words, A-G, below.

The debate over language acquisition has two main theories. B.F. Skinner believed that children learn language through a process of 17. _____. On the other hand, Noam Chomsky argued that humans possess an 18. _____ ability to understand language grammar. He pointed out that children create 19. _____ that are entirely original. Currently, researchers believe that

learning a language requires both biological factors and 20. _____ from the surrounding
21. _____.

A. imitation B. innate C. environment D. sentences E. difficulty F. input G. biology